WiP: ZeCoST: Zero-Compute Streaming Sensors using Tunnel Diodes

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Abstract

While backscatter has made wireless transmission energy-efficient, the processing pipeline including amplification, digitization, computation, remains the final energy bottleneck, especially for continuous streaming applications. To overcome this, we argue for computation-free sensing: where the processing block is eliminated entirely and raw sensor data are modulated directly onto a radio wave. Prior backscatter-based computation-free systems were fundamentally constrained by short ranges, dependence on external emitter infrastructure, and the need for power-consuming amplifiers to condition weak sensor signals. We present ZeCoST, an architecture based on tunnel diode oscillators that leverages its self-modulation and self-oscillating mixing properties to directly modulate the carrier with weak, unprocessed sensor signals, eliminating the need for amplifiers and digital stages. Consuming under 200 µW, ZeCoST achieves tens of meters of range indoors. Furthermore, ZeCoST functions as an interdomain relay, directly replicating cross-domain signals such as optical, acoustic, and electromagnetic signals onto an RF carrier to enable data reception from underwater and within a human body phantom.

CCS Concepts

• **Hardware → Sensor devices and platforms**; **Wireless devices**; Wireless integrated network sensors.

Keywords

Tunnel diodes, Backscatter communication, Zero-compute sensing, Low-power communication

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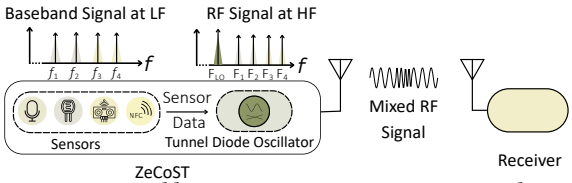


Figure 1: ZeCoST enables zero-compute streaming sensors that translate physical phenomena into radio waves with near-zero latency and sub-200 µW power consumption. A tunnel diode directly modulates a self-generated carrier with even the weakest sensor signals, eliminating amplifier, digitization, and computational processing.

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1 Introduction

A critical class of wireless embedded systems (WES), which we term *streaming sensors* [4, 8, 12], requires the real-time communication of raw sensor data to an edge device for processing. Key examples include continuous acoustic or vibration monitoring. These applications impose severe energy constraints: a typically weak µV–mV sensor signal must be amplified, digitized by an ADC, processed by a microcontroller, and transmitted, each an energy-expensive stage. Additionally with the requirement of high sampling rates, continuous streaming applications remain power-intensive, creating a significant barrier for long-term deployment.

Recent efforts address this by connecting sensors directly to a backscatter module [1, 4, 8], eliminating the computational and digitization bottleneck. However, this architecture suffers from two limitations. First, the transistor switch on a backscatter module requires a relatively high voltage to toggle; weak sensor signals must therefore still pass through an energy-expensive amplification stage. Second, backscatter mechanism is inefficient, achieving only short communication ranges and requiring precise proximity to an emitter device (ED), limiting practical adoption.

We present ZeCoST, an architecture that leverages the unique nonlinear properties of tunnel diodes to overcome

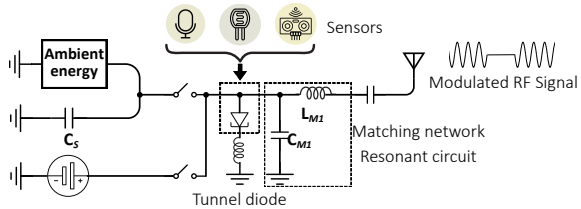


Figure 2: The TDO generates a high-frequency carrier which is modulated with the sensor baseband signals at under 200 µW. It also supports simultaneous multi-sensor upconversion.

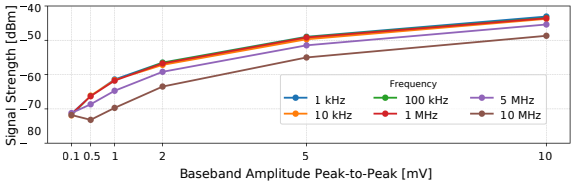


Figure 3: Variation of received signal strength with baseband amplitude across multiple signal frequencies. ZeCoST effectively mixes signals as weak as 100 µV, demonstrating its strong non-linear response and suitability for low-voltage operation.

the fundamental challenges of range and sensor interfacing in computation-free streaming sensors (Figure 1). The core of ZeCoST is a tunnel diode oscillator (TDO) that generates its own carrier locally, overcoming the need for a proximate ED. The key contribution of ZeCoST is in its direct zero-compute interface with both analog and digital sensors. We leverage the TDO's self-oscillating mixing (SoM) and self-modulation properties to directly modulate the carrier with weak, unprocessed sensor signals, eliminating the amplifier. ZeCoST also supports multiple concurrent sensors, assigning each a distinct frequency sideband.

While ZeCoST can enable numerous energy-constrained streaming applications, in this work, we focus on demonstrating its capabilities as an interdomain relay. Many emerging sensors operate in RF hostile environments (e.g., underwater or in-body) and must use optical, acoustic, or low-frequency electromagnetic (EM) signals. To bridge these sensors with wider computing infrastructure, we instantiate ZeCoST as a zero-compute relay. We demonstrate its ability to receive these diverse, inter-domain signals and replicate them onto a RF carrier, successfully enabling communication from sensors located underwater and within human body phantom.

2 Design

The core of ZeCoST is a sensor-agnostic RF frontend designed using a TDO. By biasing the tunnel diode in its Region of Negative Resistance (RNR) and pairing it with a resonant LC-tank, we construct the TDO. However, free-running TDOs suffer from frequency drift [9–11]. Prior systems stabilized the TDO through injection locking [3, 6, 10, 11], which reintroduces a dependency on an ED. Through a systematic evaluation of the TDO's operating characteristics, we identified a specific bias voltage inside the RNR where the TDO exhibits the highest carrier stability and minimum phase noise while still oscillating, producing a carrier as strong as −10 dBm. Furthermore, the inherent nonlinearity of the TDO allows it to act as a self-oscillating [10] and self-modulating [2, 5] mixer. This eliminates the need for a separate mixer or modulator blocks.

Interfacing digital sensors. A digital sensor's bitstream directly controls the TDO supply voltage, modulating the carrier with OOK in direct correspondence with the sensor output. To support multiple concurrent digital sensors, each sensor drives a dedicated low-power oscillator generating a unique baseband sinusoid (f_i); the TDO's SoM property upconverts each to a distinct RF sideband at $f_{LO} \pm f_i$, assigning every sensor its own frequency channel without analog-to-digital converters, mixers, or digital processing.

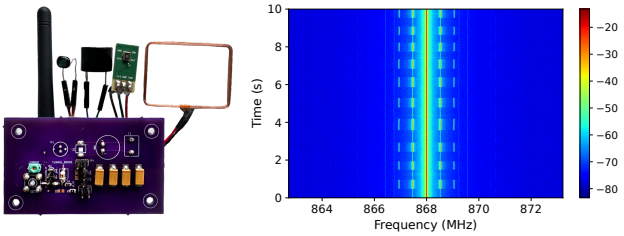
Interfacing analog sensors. Interfacing directly with analog sensors exposes the severe limitations of existing architectures. Analog sensors (e.g., photodiodes, microphones) typically generate weak µV–mV signals. Conventional pipelines condition these with power-intensive analog frontends and rely on microcontrollers for acquisition causing rapid battery depletion [12]. Backscatter-based zero-compute systems eliminate the microcontroller but rely on digital transistor switches, which are fundamentally incompatible with continuous analog waveforms and still require energy-intensive amplifiers to drive the toggle. ZeCoST overcomes both challenges. By leveraging SoM [11] and self-modulation [2] properties of TDO, the architecture directly upconverts µV-level analog signals to the RF carrier. This completely eliminates the analog frontend and microcontroller, enabling continuous, real-time analog streaming with zero processing delay.

Interdomain relay. Because the ZeCoST frontend interface is sensor-agnostic, it inherently functions as an interdomain relay. Any transducer that converts an incoming non-RF signal into a weak voltage, such as, a photodiode, an ultrasonic receiver, an inductively coupled coil, can drive the TDO, which replicates the signal onto its RF carrier without demodulation, digitization, or circuit modification.

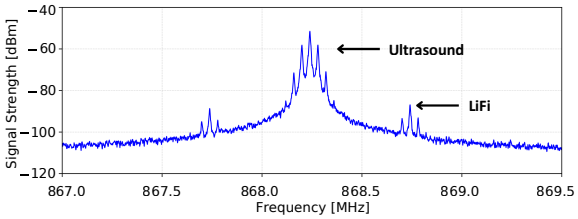
3 Preliminary Results

We evaluate ZeCoST along four axes: the weakest sensor signal it can upconvert, concurrent multi-sensor operation, RF communication range, and its feasibility as an interdomain relay. Unless stated otherwise, experiments are conducted in a multipath-rich university building using a Signal Hound BB60C spectrum analyzer to monitor the RF spectrum.

Sensitivity. To evaluate the analog interface and determine system sensitivity, the minimum sensor signal amplitude ZeCoST can successfully upconvert without amplification, we sweep baseband tones from a waveform generator. We observe clear mixed sidebands for inputs as small as 100 µV, from 1 kHz to 10 MHz (Figure 3). The sidebands grow almost



(a) ZeCoST prototype (b) Digital sensor stream



(c) Concurrent analog sensor spectrum

Figure 4: ZeCoST directly upconverts digital and analog sensor signals. (b) A 500 kHz OOK subcarrier carrying digital data drives the TDO, whose self-oscillating nonlinearity yields data-bearing sidebands at $f_c \pm f_m$ without amplification, ADC, or baseband processing. (c) RF spectrum showing concurrent ultrasound (40 kHz) and LiFi (500 kHz) sidebands from two analog sensors on a single TDO.

linearly at small amplitudes, consistent with second-order mixing. This confirms that the TDO's nonlinearity directly upconverts the weakest analog transducer outputs, eliminating the need for amplification. While baseline TDO static power dominates the energy budget to maintain the 868 MHz carrier, the dynamic load introduced by coupling external sensors is negligible. Consequently, independent of the sensors' own operating power, the total consumption of the ZeCoST front-end remains consistently under 200 µW.

Digital sensors. We validate the digital interface by feeding a 500 kHz OOK subcarrier carrying digital data to the TDO. The waterfall plot in Figure 4(b) shows the TDO's SoM upconverting the keyed subcarrier to sidebands at f_{LO}±500 kHz that toggle with the data and remain cleanly separable from the carrier over the full 10 s capture.

Multiple sensors together. We interface an ultrasonic receiver and a photodiode to a single TDO on our multisensor board (Figure 4(a)) and drive them simultaneously with a 40 kHz ultrasonic transmitter and a 500 kHz LiFi source. Both sets of upconverted sidebands appear concurrently in the RF spectrum as shown in Figure 4(c), demonstrating that the TDO can interface with multiple sensors and upconvert their emissions at the same time; the weaker optical branch is due to lower light intensity. We did not observe a reduction in sideband strength attributable to driving more than one modality concurrently. Ultimately, the theoretical limit on concurrent sensors is bounded by the LC circuit's bandwidth and the TDO's optimal stability bias voltage.

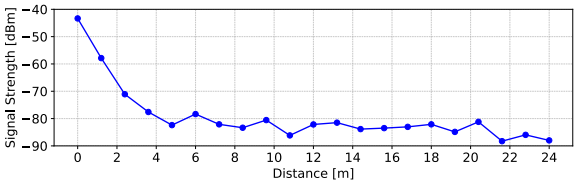


Figure 5: Indoor range evaluation for ZeCoST: in a line-of-sight yet multipath-rich environment, a detectable sideband persists up to 24 m when using a spectrum analyzer as the receiver. Receivers with higher sensitivity would proportionally extend this range.

Range. The mixed sensor sidebands remain detectable at 24 m (Figure 5) in a line-of-sight, multipath-rich university indoor environment and at 11 m in a non-line-of-sight deployment across multiple walls; the carrier itself remains detectable at 33 m. The stable signal strength observed beyond 5 m is attributed to constructive multipath reflections. Because this continuous-wave analog modulation lacks inherent multipath-resilient mechanisms, future receiver designs will require digital equalization to mitigate multipath distortion and accurately recover the original waveform.

Interdomain relay. The ultrasound and LiFi sidebands in Figure 4(c) demonstrate ZeCoST as interdomain (non-RF to RF) relay. We further prototype three applications: an underwater sensor transmitting 295 kHz ultrasound to a surface relay that replicates the signal onto radio waves up to 50 cm through water (limited by the transducer, not the relay); an on-body wearable transmitting 5 kbps to 5 Mbps OOK signals through the human body to a patch relay; and an in-body scenario using inductively coupled coils to relay signals up to 2 MHz across a 5 cm human body phantom. These experiments demonstrate the feasibility of ZeCoST for networking sensors in RF-challenged environments.

4 Conclusion and Ongoing Work

ZeCoST realizes computation-free sensing by driving a tunnel diode oscillator directly with raw transducer outputs, eliminating amplifiers, analog-to-digital converters, and microcontrollers while consuming under 200 µW. Our ongoing work targets two directions: (1) systematically characterizing concurrent multi-sensor operation and its scaling limits, and (2) integration with our parallel effort, M^2 oscillator [7]. Replacing the LC tank with M^2's high-Q SAW resonator achieves ppm-level carrier stability at 105 µW. Coupling ZeCoST's zero-compute sensor interface to this architecture will enable stable streaming, while its autodyning reception [3] provides a path to bidirectional operation.

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